Harmonic Branching Mass and Mills-Type Prime-Representing Functions below the Exponent 20/11

A mass–compactness principle for prime trees and the exponent problem of Mills, Kuipers, Niven and Wright

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Abstract

For a real number $c > 1$ let $\mathcal{W}(c)$ be the set of $A > 1$ such that $\lfloor A^{c^n} \rfloor$ is prime for every $n \geq 1$. Mills proved $\mathcal{W}(3) \neq \emptyset$ in 1947, and the admissible exponents have since been extended by Kuipers, Niven, Wright, Matomäki ($c \geq 2$), Islam ($c \geq 1.946$) and Järviniemi ($c \geq 20/11$). The published arguments are *local*: each new prime is selected inside the single short interval prescribed by the previous prime, so they require power-saving bounds for the exceptional set of short intervals that are poor in primes. We formulate a *harmonic branching problem* for trees of nested prime intervals and prove a *mass–compactness principle*: if, at every level, the harmonic mass $\sum 1/x$ carried by defective nodes is at most half of the guaranteed lower bound for the harmonic mass of that level, then the tree has an infinite branch. The principle only asks the exceptional set to beat the accumulated logarithmic losses, which grow like $\exp(O(n^2))$, while the exceptional sets coming from zero-density estimates shrink doubly exponentially along the tree. Combining it with a Selberg-type mean-square estimate with Vinogradov–Korobov savings we prove, unconditionally, that $\mathcal{W}(c)$ is uncountable, unbounded, Lebesgue-null and nowhere dense for every $c > 6/5$ using only the zero-density estimates of Ingham and Huxley, and for every $c > 15/13$ using in addition the zero-density estimate of Guth and Maynard. Under the density hypothesis the same holds for every $c > 1$. The results hold for bounded variable exponent sequences as well. The threshold $15/13$ is exactly $1/(1-\theta)$ with $\theta = 2/15$ the current almost-all exponent for the prime number theorem in short intervals. We also record an explicit chain of 22 proven primes for the exponent $5/4$ (up to 326 digits). The full conjectural range $1 < c \leq 15/13$ remains open unconditionally.

Keywords: Mills' constant; prime-representing functions; primes in almost all short intervals; zero-density estimates; König's lemma; harmonic mass.
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1 Introduction

1.1 The exponent problem

Mills [19] showed that there is a real number A such that $\lfloor A^{3^n} \rfloor$ is prime for every $n \geq 1$. The exponent 3 comes from Ingham's theorem that $[x, x + x^{5/8+\varepsilon}]$ contains primes for large x. For $c > 1$ and, more generally, for a sequence $\mathbf{c} = (c_k)_{k \geq 1}$ of reals $c_k > 1$ with $C_n = c_1 c_2 \cdots c_n$, put

$$\mathcal{W}(c) = \{A > 1 : \ \lfloor A^{c^n} \rfloor \text{ is prime for all } n \geq 1\},$$
$$\mathcal{W}(\mathbf{c}) = \{A > 1 : \ \lfloor A^{C_n} \rfloor \text{ is prime for all } n \geq 1\}.$$

The question for which c the set $\mathcal{W}(c)$ is nonempty has a long history, summarised in Table 1. Kuipers [10] treated integers $c \geq 3$, Niven [20] real $c > 8/3$ and Wright [27] $c \geq 925/348$; the theorem of Baker, Harman and Pintz on primes in intervals $[x, x + x^{0.525}]$ gives $c \geq 2.106$; Matomäki [18] reached $c \geq 2$ by working with almost all short intervals; Islam [7] reached $c \geq 1.946$ and Järviniemi [9] reached $c \geq 20/11$. Saito [23] proved that $\mathcal{W}(\mathbf{c})$ has Hausdorff dimension 1 for bounded sequences with $c_k \geq 2$. All exponents $c > 1$ are expected to be admissible, since Cramér's model predicts primes in every interval $[x, x + (\log x)^3]$.

Table 1: Admissible exponents for Mills-type prime-representing functions $\lfloor A^{c^n} \rfloor$.

Source	admissible c	input
Mills 1947 [19]	$c = 3$	Ingham: primes in $[x, x + x^{5/8}]$
Kuipers 1950 [10]	integers $c \geq 3$	same
Niven 1951 [20]	$c > 8/3$	same
Wright 1954 [27]	$c \geq 925/348 \approx 2.658$	general representing functions
via Baker–Harman–Pintz 2001 [1]	$c \geq 2.106$	primes in $[x, x + x^{0.525}]$
Matomäki 2010 [18]	$c \geq 2$	large gaps are rare [17]
Islam 2015 [7]	$c \geq 1.946$	refined exceptional sets
Järviniemi 2022 [9]	$c \geq 20/11 \approx 1.818$	sum of gaps $\geq x^{0.45}$ is $O(x^{0.63+\varepsilon})$
this paper, Theorem 1.1	$c > 6/5$	Ingham–Huxley density, VK zero-free region
this paper, Theorem 1.2	$c > 15/13 \approx 1.1538$	adds Guth–Maynard density [4]
this paper, Theorem 1.3	$c > 1$	density hypothesis

1.2 Main results

Theorem 1.1 (unconditional, classical inputs)**.** *Let* $\mathbf{c} = (c_k)_{k \geq 1}$ *satisfy* $6/5 < \inf_k c_k \leq \sup_k c_k < \infty$. *Then* $\mathcal{W}(\mathbf{c})$ *is uncountable, unbounded, of Lebesgue measure zero and nowhere dense. In*

particular, for every real $c > 6/5$ there are uncountably many $A > 1$ such that $\lfloor A^{c^n} \rfloor$ is prime for every $n \geq 1$.

Theorem 1.2 (unconditional, with the Guth–Maynard estimate)**.** *The conclusion of Theorem 1.1 holds for every sequence with $15/13 < \inf_k c_k \leq \sup_k c_k < \infty$. In particular $\mathcal{W}(c) \neq \emptyset$ for every $c > 15/13$.*

Theorem 1.3 (conditional)**.** *Assume the density hypothesis $N(\sigma, T) \ll_\varepsilon T^{2(1-\sigma)+\varepsilon}$ for $1/2 \leq \sigma \leq 1$. Then the conclusion of Theorem 1.1 holds for every sequence with $1 < \inf_k c_k \leq \sup_k c_k < \infty$.*

The exponents $6/5$ and $15/13$ are $1/(1-\theta_*)$ with $\theta_* = 1/6$ (Huxley) and $\theta_* = 2/15$ (Guth–Maynard), the known exponents for the prime number theorem in almost all short intervals [5, 4, 26]. In the range $15/13 < c < 20/11$ the theorems are new; for $c \geq 20/11$ they reprove known statements by a different method.

1.3 The new principle

Every proof in Table 1 is local. Given p_n, the next prime p_{n+1} must be chosen in $J(p_n) = [p_n^c, (p_n+1)^c - 1)$, an interval of length $\asymp X^\theta$ at height $X = p_n^c$, with $\theta = 1 - 1/c$. One needs a prime $p_{n+1} \in J(p_n)$ whose own interval $J(p_{n+1})$ is again rich in primes. Since $J(p_n)$ contains only about $X^\theta/\log X$ primes, the exceptional set of bad intervals in the whole dyadic range must be smaller than X^θ in count, i.e. one needs a *power saving*. This is what limits the method to $c \geq 20/11$ at present.

We replace the local selection by a *global* count. We start from *all* primes in a dyadic interval $[P, 2P]$, keep the whole tree of admissible continuations, and measure each level by the harmonic mass $\sum_x 1/x$. Good nodes multiply their mass by about $1/\log x$; bad nodes are discarded. The accumulated loss after n levels is $\exp(O(n^2 + n \log\log P))$, whereas the harmonic mass of *all* bad integers at level n is $\exp(-\kappa(\log Y_n)^{1/3-o(1)})$ with $\log Y_n \asymp c^n \log P$, which is doubly exponentially small in n. König's lemma then turns "every level is nonempty" into an infinite branch. No single node is ever required to have a good child. The abstract form of this is the mass–compactness principle, Theorem 2.3, which answers the harmonic branching problem, Problem 2.2. The logical structure of the paper is therefore: a new problem (Problem 2.2), a new method solving it (Theorem 2.3 and Corollary 2.5), and its application, through the estimates of Section 4, to the classical exponent problem (Section 5).

1.4 Status of the claims

Theorems 1.1 and 1.2 are unconditional. They rely on published zero-density estimates (Ingham [6], Huxley [5], Guth–Maynard [4], published in 2026), on the Vinogradov–Korobov zero-free region [8, 3] and on the explicit formula [2]; everything else is proved here. Theorem 1.3 is conditional. The problem of admissibility for all $c \in (1, 15/13]$ is *not* solved unconditionally; see Appendix B for an itemised audit. The numerical data of Section 7 are an illustration and are not used in any proof; Appendix A records a stopped side computation.

2 The harmonic branching problem and the mass–compactness principle

Definition 2.1 (branching system)**.** A *branching system* consists of a finite nonempty set $V_1 \subset \mathbb{Z}_{\geq 2}$ (the roots), for each level $n \geq 1$ a map $x \mapsto \mathrm{Ch}_n(x)$ assigning to every integer $x \geq 2$ a finite set $\mathrm{Ch}_n(x) \subset \mathbb{Z}_{\geq 2}$ of children such that $\mathrm{Ch}_n(x) \cap \mathrm{Ch}_n(x') = \emptyset$ for $x \neq x'$, and for each level a set $G_n \subset \mathbb{Z}_{\geq 2}$ of *good* integers. The levels are defined recursively by

$$V_{n+1} = \bigcup_{x \in V_n \cap G_n} \mathrm{Ch}_n(x).$$

The weight of a finite set S is $w(S) = \sum_{x\in S} 1/x$.

Problem 2.2 (harmonic branching problem)**.** Find verifiable conditions on the *local branching factors* $\lambda_n = \inf_{x\in V_n\cap G_n} w(\mathrm{Ch}_n(x))\,x$ and on the *defect masses* $\beta_n \ge w(V_n \setminus G_n)$ which guarantee an infinite branch $x_1 \in V_1$, $x_{k+1} \in \mathrm{Ch}_k(x_k)$, $x_k \in G_k$ for all $k \ge 1$, without any information on which individual nodes are good.

Theorem 2.3 (mass–compactness principle)**.** *Consider a branching system and positive numbers λ_n, nonnegative numbers β_n such that for every $n \ge 1$:*

(i) $w(\mathrm{Ch}_n(x)) \ge \lambda_n/x$ *for every* $x \in V_n \cap G_n$*;*

(ii) $w(V_n \setminus G_n) \le \beta_n$*;*

(iii) $\beta_n \le \dfrac{1}{2}\, w(V_1) \displaystyle\prod_{k=1}^{n-1} \frac{\lambda_k}{2}$.

Then $w(V_n) \ge w(V_1)\prod_{k<n}(\lambda_k/2) > 0$ for every n, and there is an infinite branch $(x_k)_{k\ge1}$ with $x_1 \in V_1$, $x_{k+1} \in \mathrm{Ch}_k(x_k)$ and $x_k \in G_k$ for all k.

Proof. Put $H_n = w(V_n)$ and $M_n = w(V_1)\prod_{k<n}(\lambda_k/2)$. We show $H_n \ge M_n$ by induction; $H_1 = M_1$. If $H_n \ge M_n$, then (ii) and (iii) give $w(V_n \cap G_n) \ge H_n - \beta_n \ge H_n - M_n/2 \ge H_n/2$. Since the sets $\mathrm{Ch}_n(x)$ are pairwise disjoint, (i) yields

$$H_{n+1} = \sum_{x\in V_n\cap G_n} w(\mathrm{Ch}_n(x)) \ge \lambda_n\, w(V_n \cap G_n) \ge \frac{\lambda_n}{2} H_n \ge M_{n+1}.$$

Hence every V_n is nonempty. Each V_n is finite, because V_1 and all $\mathrm{Ch}_n(x)$ are finite. Every $y \in V_{n+1}$ has exactly one parent $x \in V_n \cap G_n$ with $y \in \mathrm{Ch}_n(x)$, by disjointness. Thus the V_n form a locally finite forest with finitely many roots and with nodes at every depth. By König's lemma (applied after adjoining a common root) there is an infinite path $x_1, x_2, \dots$ with $x_{k+1} \in \mathrm{Ch}_k(x_k)$; a node of V_k has children only if it lies in G_k, so $x_k \in G_k$ for all k. □

Remark 2.4 (why the principle is global)**.** Condition (iii) compares the defect mass at level n with the *accumulated* mass of the whole level. It is irrelevant whether the bad nodes are concentrated below a particular node; this is the point where local arguments need power savings and Theorem 2.3 does not.

Corollary 2.5 (binary splitting)**.** *Consider a branching system with numbers λ_n, β_n satisfying* (i) *and* (ii) *of Theorem 2.3, and put $m_n = \frac{1}{3}\, w(V_1)\prod_{k=1}^{n-1}(\lambda_k/6)$. Assume that for every $n \ge 1$*

$$\textit{(iii}')\qquad \beta_n \le \tfrac{1}{2}\, m_n \qquad \textit{and} \qquad \max_{x\in V_n} \frac{1}{x} \le m_n.$$

Then there are uncountably many infinite branches $x_1 \in V_1$, $x_{k+1} \in \mathrm{Ch}_k(x_k)$, $x_k \in G_k$, and any two of them differ at some level.

Proof. A finite set S of integers with $\max_{x\in S} 1/x \le w(S)/3$ can be split into two disjoint parts of weight at least $w(S)/3$ each: order S and cut after the shortest initial segment of weight $\ge w(S)/3$; that segment has weight at most $2w(S)/3$. For $s = (s_1, s_2, \dots) \in \{0,1\}^{\mathbb{N}}$ put $U_1^s = V_1$, let V_n^s be the part of U_n^s selected by s_n, and $U_{n+1}^s = \bigcup_{x\in V_n^s\cap G_n} \mathrm{Ch}_n(x)$. Clearly $U_n^s, V_n^s \subset V_n$. We show $w(U_n^s) \ge 3m_n$ and $w(V_n^s) \ge m_n$ by induction. For $n = 1$, $w(U_1^s) = w(V_1) = 3m_1$, the split is possible because $\max 1/x \le m_1$, and $w(V_1^s) \ge m_1$. If $w(V_n^s) \ge m_n$, then by (ii) and (iii′) $w(V_n^s \cap G_n) \ge m_n - \beta_n \ge m_n/2$, so by (i) and disjointness $w(U_{n+1}^s) \ge \lambda_n m_n/2 = 3m_{n+1}$; the split is possible because $\max_{x\in V_{n+1}} 1/x \le m_{n+1}$, and $w(V_{n+1}^s) \ge m_{n+1}$. Hence every level of the tree of s is nonempty, and König's lemma gives an infinite branch $b(s)$ in it. If $s \ne s'$ first differ at index n, the n-th nodes of $b(s)$ and $b(s')$ lie in the two disjoint parts of $U_n^s = U_n^{s'}$, so $b(s) \ne b(s')$. □

3 Nested prime intervals

For $c > 1$ and an integer $x \geq 2$ put

$$J_c(x) = [x^c,\, (x+1)^c - 1), \qquad \mathcal{P}_c(x) = \mathcal{P} \cap J_c(x),$$

where $\mathcal{P}$ is the set of primes. The intervals $J_c(x)$, $x \geq 2$, are pairwise disjoint.

Lemma 3.1 (nesting)**.** *Let* $\mathbf{c} = (c_k)$ *with* $c_k > 1$*, and let* $(p_n)_{n\geq 1}$ *be integers* $p_n \geq 2$ *with* $p_{n+1} \in J_{c_{n+1}}(p_n)$ *for all* $n \geq 1$*. Then* $A = \lim_{n\to\infty} p_n^{1/C_n}$ *exists,* $A > 1$*, and* $\lfloor A^{C_n} \rfloor = p_n$ *for every* $n \geq 1$*.*

Proof. Let $a_n = p_n^{1/C_n}$ and $b_n = (p_n+1)^{1/C_n}$. From $p_{n+1} \geq p_n^{c_{n+1}}$ we get $a_{n+1} \geq a_n$, and from $p_{n+1} + 1 < (p_n+1)^{c_{n+1}}$ we get $b_{n+1} < b_n$. Hence $a_n \leq A \leq b_{n+1} < b_n$ for every n, that is $p_n \leq A^{C_n} < p_n + 1$. Finally $A \geq a_1 = p_1^{1/c_1} > 1$. □

Lemma 3.2 (interval length)**.** *For* $c > 1$ *and* $x \geq 2$ *one has* $c\,x^{c-1} - 1 \leq |J_c(x)| \leq c\,(x+1)^{c-1}$*.*

Proof. By the mean value theorem $(x+1)^c - x^c = c\,\xi^{c-1}$ with $x < \xi < x+1$, and $|J_c(x)| = (x+1)^c - x^c - 1$. □

4 Primes in almost all short intervals with Vinogradov–Korobov savings

Let $N(\sigma, T)$ denote the number of zeros $\rho = \beta + i\gamma$ of $\zeta(s)$ with $\beta \geq \sigma$ and $|\gamma| \leq T$. Write $\mathcal{L}(X) = (\log X)^{1/3}(\log\log X)^{-1/3}$.

Hypothesis 4.1 (Z(θ_1))**.** Let $0 < \theta_1 < 1$. There are $\sigma_1 \in [1/2, 1)$, $\delta_0 > 0$ and $B \geq 0$ such that, with $a_* = 2/(1-\theta_1) - \delta_0$,

(a) for every $\varepsilon > 0$: $N(\sigma, T) \ll_\varepsilon T^{a_*(1-\sigma)+\varepsilon}$ uniformly for $1/2 \leq \sigma \leq \sigma_1$, $T \geq 2$;

(b) $N(\sigma, T) \ll T^{a_*(1-\sigma)}(\log T)^B$ uniformly for $\sigma_1 \leq \sigma \leq 1$, $T \geq 2$.

We use the following classical facts.

(F1) Explicit formula [2, Ch. 17]: for $x \geq 2$ and $T \geq 2$, $\psi(x) = x - \sum_{|\gamma|<T} x^\rho/\rho + O\big(x(\log xT)^2/T + \log x\big)$.

(F2) Zero counting [2, Ch. 15]: $N(t+1) - N(t) \ll \log(|t|+2)$ for the number $N(t)$ of zeros with $0 < \gamma \leq t$.

(F3) Vinogradov–Korobov [8, Thm. 6.1], [3]: there is $c_0 > 0$ such that $\zeta(\sigma + it) \neq 0$ for $\sigma \geq 1 - c_0(\log|t|)^{-2/3}(\log\log|t|)^{-1/3}$, $|t| \geq 3$.

(F4) Ingham [6], [8, Ch. 11]: $N(\sigma, T) \ll T^{3(1-\sigma)/(2-\sigma)}(\log T)^5$ for $1/2 \leq \sigma \leq 1$.

(F5) Huxley [5], [8, Ch. 11]: $N(\sigma, T) \ll T^{3(1-\sigma)/(3\sigma-1)}(\log T)^{44}$ for $3/4 \leq \sigma \leq 1$.

(F6) Guth–Maynard [4], in the form recorded in [26, "Guth–Maynard bound"]: for each fixed $\sigma \in [7/10, 8/10]$ and every $\varepsilon > 0$, $N(\sigma, T) \ll_{\sigma,\varepsilon} T^{15(1-\sigma)/(3+5\sigma)+\varepsilon}$.

Lemma 4.2 (verification of Z(θ_1))**.** (i) Z(θ_1) *holds for every* $\theta_1 > 1/6$*, using* (F4) *and* (F5) *only.*

(ii) Z(θ_1) *holds for every* $\theta_1 > 2/15$*, using* (F4)*,* (F5) *and* (F6)*.*

(iii) *Under the density hypothesis,* $\mathrm{Z}(\theta_1)$ *holds for every* $\theta_1 > 0$.

Proof. (i) On $[1/2, 3/4]$ the exponent $3/(2-\sigma)$ of (F4) is at most $12/5$, and on $[3/4, 1]$ the exponent $3/(3\sigma-1)$ of (F5) is at most $12/5$. Hence $N(\sigma, T) \ll T^{12(1-\sigma)/5}(\log T)^{44}$ on $[1/2, 1]$. If $\theta_1 > 1/6$ then $12/5 < 2/(1-\theta_1)$; take $\sigma_1 = 1/2$ and $\delta_0 = 2/(1-\theta_1) - 12/5$ in (b).

(ii) Let $\theta_1 > 2/15$, so that $30/13 < 2/(1-\theta_1)$. On $[1/2, 7/10]$, (F4) gives exponent $3/(2-\sigma) \le 30/13$ with a logarithmic factor; on $[7/10, 4/5]$, (F6) gives $15/(3+5\sigma) \le 30/13$ with a factor T^{ε}; on $[4/5, 1]$, (F5) gives $3/(3\sigma-1) \le 15/7 < 30/13$ with a logarithmic factor. Take $\sigma_1 = 4/5$ and $\delta_0 = \frac{1}{2}(2/(1-\theta_1) - 30/13)$, so that $a_* = 30/13 + \delta_0$. Then (b) holds on $[4/5, 1]$ directly from (F5) with $B = 44$, because $15/7 < a_*$; on $[1/2, 7/10]$ the logarithmic factor in (F4) is $\ll_\varepsilon T^{\varepsilon}$, which gives (a) there. For (a) on $[7/10, 4/5]$, where (F6) is a pointwise statement, use that $N(\sigma, T)$ is nonincreasing in σ: for a mesh $7/10 = s_0 < s_1 < \cdots < s_r = 4/5$ with $s_{j+1} - s_j \le \delta_0/20$ and $\sigma \in [s_j, s_{j+1}]$, $N(\sigma, T) \le N(s_j, T) \ll T^{\frac{30}{13}(1-s_j)+\varepsilon} \le T^{\frac{30}{13}(1-\sigma)+\frac{30}{13}\cdot\frac{\delta_0}{20}+\varepsilon}$, and the extra exponent is absorbed into $\delta_0(1-\sigma)$ because $1-\sigma \ge 1/5$ and $\frac{30}{13}\cdot\frac{1}{20} < \frac{1}{5}$.

(iii) Let $\theta_1 > 0$. The density hypothesis gives (a) with exponent $2 < 2/(1-\theta_1)$ on $[1/2, 1]$. For (b) choose $\sigma_1 < 1$ with $3/(3\sigma_1 - 1) < 2/(1-\theta_1)$, which is possible because $3/(3\sigma-1) \to 3/2$ as $\sigma \to 1$, and use (F5) on $[\sigma_1, 1]$. Take δ_0 smaller than both gaps. If the density hypothesis is only available pointwise in σ, the mesh argument of (ii) gives the required uniformity. □

Proposition 4.3 (mean square with VK savings; classical, cf. [13, 14, 15, 16])**.** *Assume* $\mathrm{Z}(\theta_1)$ *and let* $\theta_1 \le \theta_2 < 1$. *There are* $\kappa > 0$ *and* X_0 *such that for* $X \ge X_0$ *and* $X^{\theta_1} \le h \le X^{\theta_2}$,

$$\int_X^{2X} \left|\psi(u+h) - \psi(u) - h\right|^2 du \le h^2 X \exp(-\kappa\, \mathcal{L}(X)).$$

Proof. Let $T = X^{1-\theta_1}\exp(2\sqrt{\log X})$; then $2 \le T \le X$ for large X. By (F1), for $u \in [X, 2X]$,

$$\psi(u+h) - \psi(u) - h = -S(u) + O\Big(\frac{X\log^2 X}{T} + \log X\Big),$$
$$S(u) = \sum_{|\gamma|<T} \frac{(u+h)^\rho - u^\rho}{\rho} = \int_u^{u+h} F(v)\, dv,$$

with $F(v) = \sum_{|\gamma|<T} v^{\rho-1}$. The error term is at most $h\exp(-\sqrt{\log X})$ for large X, since $X\log^2 X/T = X^{\theta_1}\log^2 X\, e^{-2\sqrt{\log X}}$ and $h \ge X^{\theta_1}$.

By Cauchy–Schwarz, $|S(u)|^2 \le h\int_u^{u+h} |F(v)|^2 dv$, and integrating over $u \in [X, 2X]$ (each v lies in at most h units of u-measure),

$$\int_X^{2X} |S(u)|^2 du \le h^2 \int_X^{3X} |F(v)|^2 dv = h^2 \sum_{\rho,\rho'} \int_X^{3X} v^{\rho+\overline{\rho'}-2} dv.$$

For $\rho = \beta + i\gamma$ and $\rho' = \beta' + i\gamma'$ the inner integral is at most $2X^{\beta+\beta'-1}$ in absolute value, and also at most $6X^{\beta+\beta'-1}/|\gamma-\gamma'|$ by integrating the oscillating factor. Hence it is $\le 12X^{\beta+\beta'-1}/(1+|\gamma-\gamma'|)$. With $X^{\beta+\beta'} \le (X^{2\beta} + X^{2\beta'})/2$ and symmetry,

$$\int_X^{3X} |F|^2 \le 12 \sum_{|\gamma|<T} X^{2\beta-1} \sum_{|\gamma'|<T} \frac{1}{1+|\gamma-\gamma'|} \ll (\log T)^2 \sum_{|\gamma|<T} X^{2\beta-1},$$

by (F2). Consequently

$$\int_X^{2X} |\psi(u+h) - \psi(u) - h|^2 du \ll h^2 X\Big((\log X)^2\Sigma + e^{-2\sqrt{\log X}}\Big), \qquad \Sigma = \sum_{|\gamma|<T} X^{2\beta-2}.$$

Zeros with $\beta < 1/2$ contribute at most $N(T)X^{-1} \ll T\log T/X \le X^{-\theta_1/2}$ to Σ. For $\beta \ge 1/2$, partial summation gives

$$\sum_{\beta\ge 1/2} X^{2\beta-2} = X^{-1}N(\tfrac12,T) + 2\log X \int_{1/2}^{1} X^{2\sigma-2}N(\sigma,T)\,d\sigma.$$

By (F3), $N(\sigma,T) = 0$ for $\sigma > 1-\eta$ with $\eta = c_0(\log T)^{-2/3}(\log\log T)^{-1/3}$. Write $a_*(1-\theta_1) = 2-\delta_1$ with $\delta_1 = \delta_0(1-\theta_1) > 0$. Then $T^{a_*(1-\sigma)}X^{2\sigma-2} = \exp(-(1-\sigma)[\delta_1\log X - 2a_*\sqrt{\log X}]) \le \exp(-(1-\sigma)\frac{\delta_1}{2}\log X)$ for large X. On $[\sigma_1, 1-\eta]$ hypothesis (b) bounds the integrand by $(\log X)^B\exp(-\frac{\delta_1}{2}\eta\log X)$. On $[1/2,\sigma_1]$ hypothesis (a), with $\varepsilon = (1-\sigma_1)\delta_1/4$, bounds it by $X^{-(1-\sigma_1)\delta_1/4}$. Since $\log T \asymp \log X$, we have $\eta\log X \gg \mathcal{L}(X)$, and therefore $\Sigma \ll (\log X)^{B+1}\exp(-c'\mathcal{L}(X))$ for some $c' > 0$. The proposition follows with any $\kappa < c'$. □

Definition 4.4 (good and bad integers)**.** Let $c > 1$ and $\varepsilon \in (0, 1/2]$. An integer $x \ge 2$ is (c,ε)-*good* if

$$\#\mathcal{P}_c(x) \ge (1-\varepsilon)\,\frac{|J_c(x)|}{\log((x+1)^c)},$$

and (c,ε)-*bad* otherwise.

Corollary 4.5 (bad integers are sparse)**.** *Let $1 < c_- \le c_+ < \infty$, $\varepsilon \in (0,1/2]$, and assume* $\mathrm{Z}(\theta_1)$ *for some $\theta_1 < 1 - 1/c_-$. There are $\kappa' > 0$ and Y_0 such that for $Y \ge Y_0$ and every $c \in [c_-, c_+]$,*

$$\#\{x \in [Y, 2Y) : \ x \text{ is } (c,\varepsilon)\text{-bad}\} \le Y\exp(-\kappa'\mathcal{L}(Y)).$$

Proof. Put $\theta = 1 - 1/c \in [1-1/c_-, 1-1/c_+]$ and $K = \lceil 100c_+/\varepsilon\rceil$. Split $[Y,2Y)$ into K blocks $[y_i, y_{i+1})$ with $y_i = Y(1+i/K)$. For x in block i, Lemma 3.2 gives $c\,y_i^{c-1} - 1 \le |J_c(x)| \le c\,y_i^{c-1}(1+2/K)^{c-1}$ for Y large, hence $c\,y_i^{c-1} \ge (1-\varepsilon/50)|J_c(x)|$. Put $h_i = (1-\varepsilon/4)(c\,y_i^{c-1}-1)$, so that $(1-0.3\varepsilon)|J_c(x)| \le h_i \le (1-\varepsilon/4)|J_c(x)|$ for large Y; the upper bound holds because $c\,y_i^{c-1} - 1 \le c\,x^{c-1} - 1 \le |J_c(x)|$.

Let x be bad and $u \in U(x) = [x^c, x^c + \frac{\varepsilon}{4}|J_c(x)|]$. Then $(u, u+h_i] \subset [x^c, (x+1)^c - 1]$, so the primes in $(u, u+h_i]$ are those of $\mathcal{P}_c(x)$ plus possibly the endpoint $(x+1)^c - 1$. Prime powers p^k with $k \ge 2$ in an interval of length h_i at height $\asymp Y^c$ number $O(Y^{c/2-1} + \log Y)$. Hence

$$\psi(u+h_i) - \psi(u) \le \log((x+1)^c)\,(\#\mathcal{P}_c(x)+1) + O\big((Y^{c/2-1}+\log Y)\log Y\big) \le (1-\varepsilon)|J_c(x)| + o(\varepsilon h_i),$$

and therefore $\psi(u+h_i) - \psi(u) - h_i \le -0.7\varepsilon|J_c(x)| + o(\varepsilon h_i) \le -\frac{\varepsilon}{2}h_i$. The sets $U(x)$ are pairwise disjoint and have measure $\frac{\varepsilon}{4}|J_c(x)| \ge \frac{\varepsilon}{4}h_i$; they lie in $[Y^c, (2Y+1)^c]$, which is covered by at most $2c_+ + 2$ dyadic intervals $[X, 2X]$ with $Y^c \le X \le 3^{c_+}Y^c$. For each of them $X^{\theta_1} \le h_i \le X^{\theta_2}$ with $\theta_2 = 1 - 1/(2c_+)$, which exceeds every $\theta \le 1 - 1/c_+$, once Y is large; Proposition 4.3 is applied with this fixed pair (θ_1, θ_2). By Chebyshev's inequality and Proposition 4.3, the measure of $\{u \in [X,2X] : |\psi(u+h_i) - \psi(u) - h_i| \ge \frac{\varepsilon}{2}h_i\}$ is at most $4\varepsilon^{-2}X\exp(-\kappa\mathcal{L}(X))$. Hence the number of bad x in block i is

$$\le \frac{4}{\varepsilon h_i}\,(2c_+ + 2)\,4\varepsilon^{-2}\,3^{c_+}Y^c\exp(-\kappa\mathcal{L}(Y^c)) \ll_{c_+,\varepsilon} Y\exp(-\kappa\mathcal{L}(Y^c)),$$

because $h_i \gg Y^{c-1}$. Summing over the K blocks and using $\mathcal{L}(Y^c) \ge \mathcal{L}(Y)$ for large Y gives the claim for any $\kappa' < \kappa$ and large Y. □

5 Proof of the main theorems

Let $\mathbf{c} = (c_k)$ with $c_- = \inf c_k$ and $c_+ = \sup c_k$, and let $\theta_* \in \{1/6,\ 2/15,\ 0\}$ be the threshold corresponding to Theorem 1.1, Theorem 1.2 and Theorem 1.3, so that the assumption is $c_- > 1/(1-\theta_*)$, i.e. $1 - 1/c_- > \theta_*$. Fix θ_1 with $\theta_* < \theta_1 < 1 - 1/c_-$; by Lemma 4.2, $\mathrm{Z}(\theta_1)$ holds (unconditionally in the first two cases). Fix $\varepsilon = 1/4$.

The branching system. Let P be large. Put $V_1 = \mathcal{P} \cap [P, 2P]$, $\mathrm{Ch}_n(x) = \mathcal{P}_{c_{n+1}}(x)$ and $G_n = \{x : \ x \text{ is } (c_{n+1}, \varepsilon)\text{-good}\}$. The child sets of one level are disjoint because the intervals $J_{c_{n+1}}(x)$ are. Put $D_n = c_2 c_3 \cdots c_n$ ($D_1 = 1$), $M_1 = 2P$ and $M_{n+1} = (M_n + 1)^{c_{n+1}}$. By induction $V_n \subset [P^{D_n}, M_n]$. Writing $\log M_n = D_n a_n$ and using $\log(M_n + 1) \le \log M_n + 1/M_n$ we get $a_{n+1} \le a_n + 1/M_n \le a_n + P^{-D_n}$; since $D_n \ge c_-^{n-1} \ge 1 + (n-1)(c_- - 1)$, this gives $a_n \le \log(2P) + 2/P \le \log(3P)$ for large P. Hence

$$V_n \subset R_n = \left[P^{D_n}, (3P)^{D_n}\right].$$

Local branching factor. Let $x \in V_n \cap G_n$ and $c = c_{n+1}$. Every child y satisfies $y < (x+1)^c$, and by goodness and Lemma 3.2,

$$w(\mathrm{Ch}_n(x)) \ge \frac{\#\mathcal{P}_c(x)}{(x+1)^c} \ge \frac{(1-\varepsilon)(c\,x^{c-1} - 1)}{c \log(x+1)\,(x+1)^c} \ge \frac{1 - 2\varepsilon}{x \log(x+1)} \ge \frac{\lambda_n}{x}, \qquad \lambda_n = \frac{1}{2D_n \log(4P)},$$

for P large, since $\log(x+1) \le D_n \log(4P)$ on R_n.

Defect mass. By Corollary 4.5, applied to dyadic blocks $[Y, 2Y)$ covering R_n (at most $2D_n + 2$ of them, all with $Y \ge P^{D_n}$),

$$w(V_n \setminus G_n) \le \sum_{\text{blocks}} \frac{1}{Y}\, Y e^{-\kappa' \mathcal{L}(Y)} \le (2D_n + 2) \exp\bigl(-\kappa' \mathcal{L}(P^{D_n})\bigr) =: \beta_n.$$

Initial mass. By the prime number theorem, $w(V_1) \ge (\pi(2P) - \pi(P))/(2P) \ge 1/(4 \log P)$ for large P.

Verification of (iii′). We have $c_-^{n-1} \le D_n \le c_+^{n-1}$. Write $m = n - 1$ and $\ell = \log P$. Then, with m_n as in Corollary 2.5,

$$\tfrac{1}{2} m_n = \tfrac{1}{6} w(V_1) \prod_{k<n} \frac{\lambda_k}{6} \ge \frac{1}{24\ell} \left(12 \log 4P\right)^{-m} c_+^{-m(m-1)/2},$$

$$\beta_n \le (2c_+^m + 2) \exp\Bigl(-\kappa' \frac{(c_-^m \ell)^{1/3}}{(\log(c_+^m \ell))^{1/3}}\Bigr).$$

Taking logarithms, it suffices that

$$\kappa' \frac{c_-^{m/3} \ell^{1/3}}{(m \log c_+ + \log \ell)^{1/3}} \ge \log(2c_+^m + 2) + \log(24\ell) + m \log(12 \log 4P) + \tfrac{m(m-1)}{2} \log c_+$$

for all $m \ge 0$. The right side is $\le C(1 + m^2)(1 + \log \ell)$ with $C = C(c_+)$, and the left side is $\ge \kappa' c_-^{m/3} \ell^{1/3} (m+1)^{-1/3} (\log c_+ + \log \ell)^{-1/3}$. Since $\mu = \inf_{m \ge 0} c_-^{m/3} (m+1)^{-7/3} > 0$, the inequality holds for all m as soon as $\kappa' \mu\, \ell^{1/3} \ge 2C(1 + \log \ell)(\log c_+ + \log \ell)^{1/3}$, which is true for large ℓ. The second condition in (iii′) holds because $\max_{x \in V_n} 1/x \le P^{-D_n} \le P^{-c_-^m} = e^{-c_-^m \ell}$, whereas $m_n \ge e^{-C'(1+m^2)(1+\log \ell)}$; for large ℓ the first is smaller for every $m \ge 0$.

Conclusion. Corollary 2.5 gives uncountably many infinite branches (p_n) with $p_{n+1} \in J_{c_{n+1}}(p_n)$, any two of which differ at some level. By Lemma 3.1 each branch yields $A \in \mathcal{W}(\mathbf{c})$ with $\lfloor A^{C_n} \rfloor = p_n$, so distinct branches yield distinct A. Thus $\mathcal{W}(\mathbf{c})$ is uncountable. Since $A^{c_1} \in [P, 2P + 1)$ and P can be taken arbitrarily large, $\mathcal{W}(\mathbf{c})$ is unbounded.

Remark 5.1 (size of the constants)**.** The constants A produced here are large: $A^{c_1} \geq P$ with P beyond the range where the estimates of Section 4 take effect. This is unavoidable for small c in general: for $A < 2$ the number $\lfloor A^c \rfloor$ is tiny and small cases can fail; for instance the comment to OEIS A243358 [21] records that for $A < 2$ and $c < 1.2209864\ldots$ the sequence $\lfloor A^{c^n} \rfloor$ must contain composite terms. No such restriction applies to large A. The threshold P_0 is effective in principle (all analytic inputs have effective constants), but the constant A itself is obtained through König's lemma; unlike Mills' greedy construction, the proof gives no algorithm with a known running time for computing an element of $\mathcal{W}(\mathbf{c})$.

Measure and category. Let $1 < a < b$ and $W_n = \{A \in [a,b] : \lfloor A^{C_k} \rfloor \in \mathcal{P} \ (k \leq n)\}$. Each $A \in W_n$ lies in an interval $[p^{1/C_n}, (p+1)^{1/C_n})$ with $p \in \mathcal{P}$, $a^{C_n} \leq p+1$, $p \leq b^{C_n}$, whose length is $\leq p^{1/C_n}/(C_n p) \leq b/(C_n p)$. Hence, by Mertens' theorem,

$$\operatorname{meas} W_n \leq \frac{b}{C_n} \sum_{a^{C_n}-1 \leq p \leq b^{C_n}} \frac{1}{p} \leq \frac{b}{C_n}\Big(\log \frac{\log b}{\log a} + o(1)\Big) \longrightarrow 0.$$

The closure of W_n consists of W_n and finitely many endpoints, so $\overline{\mathcal{W}(\mathbf{c}) \cap [a,b]} \subset \bigcap_n \overline{W_n}$ has measure zero. A closed null set has empty interior; hence $\mathcal{W}(\mathbf{c})$ is Lebesgue-null and nowhere dense. This proves Theorems 1.1 to 1.3. □

5.1 The weakest input: a transference theorem

The proof above uses Proposition 4.3 only through Corollary 4.5, and it needs far less than the Vinogradov–Korobov saving. The next theorem isolates what is actually required: a lower bound of the right order of magnitude (not an asymptotic) in almost all of the intervals $J_c(x)$, with an exceptional density that beats every fixed power of log by a quasi-polynomial factor.

Theorem 5.2 (transference)**.** *Let $1 < c_- \leq c_+ < \infty$ and $\delta \in (0,1]$. Suppose there are $C > \log c_+/(2(\log c_-)^2)$ and Y_0 such that for all $Y \geq Y_0$ and all $c \in [c_-, c_+]$*

$$(S) \qquad \#\Big\{x \in [Y, 2Y) : \ \#\mathcal{P}_c(x) < \delta \, \frac{|J_c(x)|}{\log((x+1)^c)}\Big\} \leq Y \exp(-C(\log\log Y)^2).$$

Then for every sequence $\mathbf{c}$ with values in $[c_-, c_+]$ the set $\mathcal{W}(\mathbf{c})$ is uncountable, unbounded, Lebesgue-null and nowhere dense.

Proof. Run the construction of this section with G_n the set of integers satisfying the inequality opposite to (S) for $c = c_{n+1}$. The local branching factor becomes $\lambda_n = \delta/(2D_n \log 4P)$, and the defect mass is $\beta_n \leq (2D_n + 2)\exp(-C(\log D_n + \log \ell)^2)$ with $\ell = \log P$. With $m = n-1$ and $\log D_n \geq m \log c_-$, condition (iii′) of Corollary 2.5 follows from

$$[C \log^2 c_- - \tfrac{1}{2}\log c_+]m^2 + [2C \log c_- \log \ell - \log c_+ - \log(24\ell/\delta)]m + [C \log^2 \ell - 2 - \log(24\ell)] \geq 0,$$

which holds for all $m \geq 0$ once ℓ is large, because each bracket is then nonnegative. The weight condition in (iii′) and the remaining steps are as before. □

Corollary 5.3. *Corollary 4.5 gives (S) with $\delta = \frac{1}{2}$ and the much stronger bound $Y \exp(-\kappa' \mathcal{L}(Y))$ whenever $c_- > A^*/2$, where A^* is a uniform zero-density exponent available with logarithmic losses near $\sigma = 1$ (Lemma 4.2). This recovers Theorems 1.1 to 1.3. Any future proof of (S) for some $\theta = 1 - 1/c$ below $2/15$, for instance a quantitative form of a sieve lower bound for primes in almost all short intervals, would immediately extend the theorems to that c.*

Remark 5.4 (how far towards $c = 1$)**.** For constant c the condition on C reads $C > 1/(2\log c)$, and the exceptional density needed is $\exp(-C(\log\log Y)^2) = (\log Y)^{-C\log\log Y}$: super-polylogarithmic, but far weaker than the Vinogradov–Korobov saving. As $c \downarrow 1$ the constant

C grows like $1/(2(c-1))$, but the saving stays of quasi-polylogarithmic type; what becomes hard is the interval length $X^{1-1/c}$. The obstruction to $c \to 1$ is therefore entirely the lack of almost-all prime information for intervals $[x, x+x^\theta]$ with small θ: lower bounds of the right order are known in almost all such intervals only for $\theta > 1/22$ [12], and asymptotics only for $\theta > 2/15$. Reaching every $c > 1$ by this method requires (S) for every $\theta > 0$, which follows from the density hypothesis and is otherwise open.

6 Local versus global: where the exponent comes from

Table 2: What each method needs from the exceptional set $E(y)$ of integers $x \in [y, 2y)$ whose interval $J_c(x)$ (length $\asymp y^{c-1} = (y^c)^\theta$, $\theta = 1 - 1/c$) is poor in primes.

Method	requirement	present range
local selection [18, 9]	$\#E(y) \le y^{\theta-\delta}$, i.e. fewer bad integers than primes in one interval $J_c(p)$, $p \asymp y^{1/c}$	$c \ge 20/11$
global mass (Theorem 2.3)	$\#E(y) \le y \exp(-C(\log\log y)^2)$ for a suitable C; VK savings suffice	$c > 15/13$

The local method must find a good child inside one interval $J_c(p_n)$; if $p_{n+1} \asymp y$, this interval contains about $y^\theta/\log y$ primes, while the bad integers are counted in the whole range $[y, 2y)$. It therefore needs a power saving of size $y^{1-\theta} = y^{1/c}$ in the exceptional set. Results of this type come from large-value estimates for Dirichlet polynomials and from sieve methods; with the bound of Järviniemi [9] they reach $\theta = 0.45$, i.e. $c = 20/11$.

The global method compares the bad mass with the mass of the whole level, which is only $\exp(O(n^2 + n\log\log P))$ smaller than the initial mass. The bad mass at level n is $\exp(-\kappa(\log Y_n)^{1/3-o(1)})$ with $\log Y_n \ge c_-^{n-1}\log P$. Any saving of the form $\exp(-(\log Y)^\delta)$ with fixed $\delta > 0$ would suffice; polylogarithmic savings would not, because the accumulated loss $\prod_{k<n}\log Y_k$ grows faster than every fixed power of $\log Y_n$ (for constant exponent c it equals $(\log Y_n)^{(1/2+o(1))\log\log Y_n/\log c}$ as $n \to \infty$). This explains the thresholds:

$$c > \frac{1}{1-\theta_{\text{PNT-AA}}}, \qquad \theta_{\text{PNT-AA}} = \tfrac{1}{6}\ \text{(Huxley)},\ \tfrac{2}{15}\ \text{(Guth–Maynard)},\ 0\ \text{(density hypothesis)}.$$

Equivalently, writing $A^* = \sup_{1/2\le\sigma\le 1} A(\sigma)$ for the best uniform zero-density exponent, the zero-density route gives exactly

$$\mathcal{W}(c) \ne \emptyset \qquad \text{for every } c > A^*/2,$$

since $1/(1-(1-2/A^*)) = A^*/2$: $A^* = 12/5$ gives $6/5$, $A^* = 30/13$ gives $15/13$, and $A^* = 2$ (the density hypothesis) gives every $c > 1$. Theorem 5.2 below isolates the weakest input the method needs. Below $c = 15/13$ the only available almost-all results for intervals $[x, x+x^\theta]$, $\theta \le 2/15$, are sieve lower bounds, down to $\theta > 1/22$ [11, 12]. Their exceptional sets have not, to our knowledge, been quantified beyond $o(X)$. If they satisfy hypothesis (S) of Theorem 5.2, the method gives every $c > 22/21 \approx 1.0476$, but not $c \to 1$: for $1 - 1/c \le 1/22$ not even the existence of a prime in almost all intervals $[x, x+x^{1-1/c}]$ is known.

Remark 6.1 (relation to Mills' constant)**.** The theorems concern existence and do not identify any specific constant. In particular they say nothing about the least element of $\mathcal{W}(3)$ (Mills' constant), whose irrationality was proved by Saito [24] and whose transcendence is open [25].

7 A certified chain for the exponent 5/4

The exponent $c = 5/4$ lies in the new range $(15/13, 20/11)$; it is covered by Theorem 1.1 since $5/4 > 6/5$. For $c = 5/4$ the interval condition is purely integral:

$$m \in J_{5/4}(x) \iff m^4 \geq x^5 \text{ and } (m+1)^4 < (x+1)^5.$$

Starting from $p_1 = 1009$ and taking each time the least prime of $J_{5/4}(p_n)$ gives 22 primes $p_1, \ldots, p_{22}$, each proven prime by `isprime` in PARI/GP 2.15 [22], with all nesting conditions checked in exact integer arithmetic (Listing 1). The first terms are

$$1009,\ 5689,\ 49409,\ 736657,\ 21581503,\ 1470964273,\ 288073066607,\ 211046580457399,\ \ldots$$

and the digit counts are $4, 4, 5, 6, 8, 10, 12, 15, 18, 23, 28, 35, 44, 55, 69, 86, 107, 134, 167, 209, 261, 326$. The offsets $p_{n+1} - \lceil p_n^{5/4} \rceil$ never exceed 954, while $|J_{5/4}(p_n)| \approx 1.25\, p_n^{1/4}$; for instance $|J_{5/4}(p_{21})| \approx 10^{65}$. Any A with $\lfloor A^{(5/4)^n} \rfloor = p_n$ for $n \leq 22$ satisfies

$$A = 253.064887471541765088407431583113669011549011540467458584789490728820 87\ldots$$

(to the precision fixed by p_{22}). This finite computation illustrates the nested-interval mechanism; it is not a proof that this greedy sequence continues forever, and Theorem 1.1 does not assert that.

A A stopped side computation: Proth candidates with 30,103 digits

Independently of the theorems, we searched for a prime larger than the 18,886-digit Pocklington prime of the author's earlier computations, in the form $N_k = k \cdot 2^{100000} + 1$ with odd $k \leq 2 \cdot 10^6$, where Proth's theorem certifies primality by a single congruence $a^{(N_k-1)/2} \equiv -1 \pmod{N_k}$. Sieving by all primes up to 10^{10} (Listing 4) left 48,707 candidates. The test of Listing 3 reduces modulo $k2^n + 1$ without division (writing $t = H2^n + L$ and $Q = \lfloor H/k \rfloor$, one has $t \equiv t - QN_k$) and was validated against `gmpy2` on smaller exponents. Each test took about 17 s on one core.

The run was stopped by the time limit of the computing environment after 1,639 candidates ($k \leq 68{,}751$), all composite; no prime was found. Heuristically a sieved candidate is prime with probability about $e^\gamma \log(10^{10}) / \log N_k \approx 5.9 \cdot 10^{-4}$, so the expected number of tests before the first prime is about 1,700, and the observed outcome has probability about $e^{-0.97} \approx 0.38$. The logs of the tested candidates and the list of the 47,068 untested ones are distributed with the source files, so that the search can be resumed without repetition. No claim is made on the basis of this computation.

B Audit against the author's success criteria

The criteria below were fixed in advance by the author. Each verdict is stated as strictly as we can.

In plain terms: the paper resolves the Mills exponent problem on the new range $15/13 < c < 20/11$, shows (Theorem 5.2, Remark 5.4) exactly which almost-all input would be needed to go further, and does not resolve the problem for $1 < c \leq 15/13$. With present knowledge no almost-all method can go below $c = 22/21$.

C Verification code

Listing 1: PARI/GP script for the chain of Section 7.

```
\\ Greedy chain for c = 5/4.  J(x) = { m : m^4 >= x^5 and (m+1)^4 < (x+1)^5 }.
\\ Each term is proven prime by isprime() (PARI/GP primality proof).
default(parisize, 10^8);
{
p = 1009; L = List([p]);
for (n = 1, 21,
  lo = sqrtnint(p^5, 4); if (lo^4 < p^5, lo++);
  m = nextprime(lo);
  if (!((m+1)^4 < (p+1)^5), error("stuck at n=", n));
  if (!isprime(m), error("not proven prime"));
  listput(L, m); p = m);
}
{
for (i = 1, #L,
  q = L[i];
  if (i == 1, off = 0, a = L[i-1]; r = sqrtnint(a^5, 4); if (r^4 < a^5, r++); off = q - r);
  printf("%d  digits=%d  offset=%d\n", i, #digits(q), off));
}
{
ok = 1;
for (i = 1, #L-1, a = L[i]; b = L[i+1]; if (!(b^4 >= a^5 && (b+1)^4 < (a+1)^5), ok = 0));
print("nesting ok: ", ok);
}
write("chain54.txt", Vec(L));
\p 400
{
N = #L; A = exp(log(L[N] + 1/2) * (4/5)^N);
print("A approx = ", A);
bad = 0; for (i = 1, N, if (floor(A^((5/4)^i)) != L[i], bad++));
print("floor mismatches: ", bad);
}
quit;
```

Listing 2: Independent Proth certificate check (Python 3 with gmpy2), for use if the search is resumed.

```
#!/usr/bin/env python3
"""Independent check of a Proth certificate N = k*2^n + 1 (k odd, k < 2^n).

Proth's theorem: if a^((N-1)/2) == -1 (mod N) for some integer a, then N is prime.
Usage: python3 verify_proth.py k n a
"""
import sys, hashlib, time
import gmpy2
from gmpy2 import mpz

def main():
    k, n, a = (int(x) for x in sys.argv[1:4])
    assert k % 2 == 1 and k > 0, "k must be odd and positive"
    assert k < 2**n, "Proth's theorem needs k < 2^n"
    N = mpz(k) * (mpz(1) << n) + 1
    t0 = time.time()
    r = gmpy2.powmod(a, (N - 1) // 2, N)
    dt = time.time() - t0
    digits = len(str(N))
    print(f"N = {k}*2^{n}+1 has {digits} decimal digits and {N.bit_length()} bits")
    print(f"Jacobi({a}, N) = {gmpy2.jacobi(a, N)}")
    print(f"a^((N-1)/2) mod N == N-1 : {r == N - 1}   ({dt:.1f} s)")
    print("SHA-256 of decimal expansion:", hashlib.sha256(str(N).encode()).hexdigest())
    if r == N - 1:
        print("PROVEN PRIME by Proth's theorem.")
    else:
```

```
        print("certificate FAILED")
        sys.exit(1)

if __name__ == "__main__":
    main()
```

Listing 3: Search program used to find the Proth prime (GMP; reduction modulo $k2^n+1$ without division).

```
// Proth test for N = k*2^n+1 with fast reduction. Usage: proth n listfile slice nslices outfile [maxsteps]
#include <gmp.h>
#include <stdio.h>
#include <stdlib.h>
#include <time.h>
static void redmod(mpz_t t, const mpz_t N, unsigned long k, unsigned long n, mpz_t Q, mpz_t u) {
  mpz_tdiv_q_2exp(Q, t, n);
  mpz_tdiv_q_ui(Q, Q, k);
  mpz_mul_ui(u, Q, k);
  mpz_mul_2exp(u, u, n);
  mpz_add(u, u, Q);
  mpz_sub(t, t, u);
  while (mpz_sgn(t) < 0) mpz_add(t, t, N);
  while (mpz_cmp(t, N) >= 0) mpz_sub(t, t, N);
}
// returns 1 if a^((N-1)/2) == -1, 0 if == +1 (inconclusive), -1 otherwise (composite)
static int proth_test(unsigned long k, unsigned long n, unsigned long a, long maxsteps){
  mpz_t N,x,t,Q,u; mpz_inits(N,x,t,Q,u,NULL);
  mpz_set_ui(N,k); mpz_mul_2exp(N,N,n); mpz_add_ui(N,N,1);
  mpz_set_ui(x,a); mpz_powm_ui(x,x,k,N);        // x = a^k mod N
  long steps = (maxsteps>0 && maxsteps < (long)n-1) ? maxsteps : (long)n-1;
  for (long i=0;i<steps;i++){ mpz_mul(t,x,x); redmod(t,N,k,n,Q,u); mpz_swap(x,t); }
  int res;
  if (maxsteps>0 && steps < (long)n-1) { res = 2; gmp_printf("%Zx\n", x); }
  else { mpz_add_ui(t,x,1); if (mpz_cmp(t,N)==0) res=1; else if (mpz_cmp_ui(x,1)==0) res=0; else res=-1; }
  mpz_clears(N,x,t,Q,u,NULL);
  return res;
}
static unsigned long pick_base(unsigned long k, unsigned long n){
  mpz_t N; mpz_init(N); mpz_set_ui(N,k); mpz_mul_2exp(N,N,n); mpz_add_ui(N,N,1);
  unsigned long ps[]={3,5,7,11,13,17,19,23,29,31,37,41,43,47,53,59,61,67,71,73,79,83,89,97};
  unsigned long a=0;
  for (int i=0;i<24;i++){ if (mpz_ui_kronecker(ps[i],N)==-1){ a=ps[i]; break; } }
  mpz_clear(N); return a;
}
int main(int argc,char**argv){
  unsigned long n=strtoul(argv[1],0,10);
  FILE*f=fopen(argv[2],"r"); int slice=atoi(argv[3]), ns=atoi(argv[4]);
  FILE*o=fopen(argv[5],"a"); long maxsteps = argc>6? atol(argv[6]) : 0;
  unsigned long k; long idx=0;
  while (fscanf(f,"%lu",&k)==1){
    if ((idx++ % ns)!=slice) continue;
    unsigned long a=pick_base(k,n);
    if (!a){ fprintf(o,"k=%lu nobase\n",k); fflush(o); continue; }
    time_t t0=time(0);
    int r=proth_test(k,n,a,maxsteps);
    fprintf(o,"k=%lu a=%lu res=%d secs=%ld\n",k,a,r,(long)(time(0)-t0)); fflush(o);
    if (r==1){ fprintf(o,"PRIME k=%lu n=%lu base=%lu\n",k,n,a); fflush(o); break; }
    if (maxsteps>0) break;
  }
  return 0;
```

```
}
```

Listing 4: Sieve used to select the Proth candidates.

```
// Sieve odd k in [1,K] for N=k*2^n+1 against primes l in [3,B]. Output surviving k.
#include <stdio.h>
#include <stdlib.h>
#include <stdint.h>
#include <string.h>
#include <math.h>
static uint64_t mulmod(uint64_t a,uint64_t b,uint64_t m){return (unsigned __int128)a*b%m;}
static uint64_t powmod(uint64_t a,uint64_t e,uint64_t m){uint64_t r=1%m;a%=m;while(e){if(e&1)
    r=mulmod(r,a,m);a=mulmod(a,a,m);e>>=1;}return r;}
static int64_t inv(int64_t a,int64_t m){int64_t g=m,x=0,x1=1,a1=a;while(a1){int64_t q=g/a1,t=
    g-q*a1;g=a1;a1=t;t=x-q*x1;x=x1;x1=t;}if(x<0)x+=m;return x;}
int main(int argc,char**argv){
  uint64_t n=strtoull(argv[1],0,10), K=strtoull(argv[2],0,10), B=strtoull(argv[3],0,10);
  uint8_t *alive=malloc(K+1); memset(alive,1,K+1);
  for(uint64_t k=0;k<=K;k+=2) alive[k]=0; // even k
  // segmented sieve of Eratosthenes for primes up to B
  uint64_t sq=(uint64_t)sqrt((double)B)+1;
  uint8_t *small=calloc(sq+1,1); uint64_t *sp=malloc(sizeof(uint64_t)*sq); size_t ns=0;
  for(uint64_t i=2;i<=sq;i++){ if(!small[i]){ sp[ns++]=i; for(uint64_t j=i*i;j<=sq;j+=i)
    small[j]=1; } }
  const uint64_t SEG=1<<24; uint8_t *seg=malloc(SEG);
  uint64_t cnt=0;
  for(uint64_t lo=3; lo<=B; lo+=SEG){
    uint64_t hi=lo+SEG-1; if(hi>B) hi=B;
    memset(seg,0,SEG);
    for(size_t t=0;t<ns;t++){ uint64_t p=sp[t]; if(p*p>hi) break; uint64_t s=((lo+p-1)/p)*p;
    if(s<p*p) s=p*p; for(uint64_t j=s;j<=hi;j+=p) seg[j-lo]=1; }
    for(uint64_t l=lo;l<=hi;l++){ if(seg[l-lo]) continue; if(l==2) continue;
      // N = k*2^n+1 = 0 mod l  <=> k = -inv(2^n) mod l
      uint64_t t2=powmod(2,n,l); uint64_t r=(l-(uint64_t)inv((int64_t)t2,(int64_t)l))%l;
      for(uint64_t k=r;k<=K;k+=l){ if(k==0) continue; alive[k]=0; }
      cnt++;
    }
  }
  uint64_t surv=0; for(uint64_t k=1;k<=K;k++) if(alive[k]){ printf("%llu\n",(unsigned long
    long)k); surv++; }
  fprintf(stderr,"primes sieved %llu survivors %llu\n",(unsigned long long)cnt,(unsigned long
     long)surv);
  return 0;
}
```

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Criterion	Assessment	Verdict
G1 prior named problem	The exponent problem for Mills-type prime-representing functions, posed implicitly by Mills (1947) and studied explicitly by Kuipers, Niven, Wright, Matomäki, Islam and Järviniemi; the statement "$\exists A$: $\lfloor A^{c^n} \rfloor$ prime $\forall n$" predates this work.	met
G2 problem unchanged	For each fixed c we prove the original statement, without modified constants or reformulation.	met
G3 actually open	For $c < 20/11$ no proof was available in the literature we could find ([9] and the exponent database [26] list 20/11 and 2/15 as current records).	met for $c \in (15/13, 20/11)$
G4 unconditional and complete	Theorems 1.1 and 1.2: complete unconditional proofs, all quantifiers closed, for every $c > 6/5$, resp. $c > 15/13$. The full conjecture (all $c > 1$) is proved only under the density hypothesis.	met on $(15/13, \infty)$; **not** met on $(1, 15/13]$
G5 not a restatement	The operative new ingredient is a quantitative non-vanishing estimate for the harmonic branching mass (Theorem 2.3) combined with VK-saving exceptional-set bounds; no identity or change of variables carries the proof.	met
G6 no prior identical solution	We found no earlier global (mass–compactness) argument for prime-representing functions and no result below 20/11; absence in our search does not prove priority.	no counterexample found